

Likelihood functions for the single bin QPAD (SQPAD) approach

In the following we describe the likelihood and composite likelihood functions used in implementing the SQPAD method and its comparison with the complete information estimator. We use the following notation. This is the same as presented in Section 2 of the main text.

1 Notation and assumptions

- S : Total number of survey locations
- y_s : Observed count or occupancy status at the s -th location where $s = 1, 2, \dots, S$
- t_s : Survey duration for the s -th location where $s = 1, 2, \dots, S$
- r_s : Radius of the spatial area that is surveyed at the s -th location where $s = 1, 2, \dots, S$
- t_{sj} : First time the j -th individual at the s -th location was perceived where $j = 1, 2, \dots, y_s$
- r_{sj} : Distance the j -th individual at the s -th location when it was perceived for the first time where $j = 1, 2, \dots, y_s$

Assumptions:

1. Individuals are distributed over space according to a Homogeneous Poisson process with density D .
2. Individuals emanate cues according to a Time-Homogeneous Poisson Process with intensity δ .
3. Individuals emanate cues independent of each other.
4. Cues are perceived depending on the distance from the observer according to a function $v(r; \varphi)$ such that $v(0, \varphi) = 1$ and $0 < v(r; \varphi) < 1$ for all $r > 0$.

1.1 Distance-independent perception of cues

We start with a simpler case where the cues are detected with probability one, independent of the distance to the observer. Removal sampling also makes this assumption.

- Let $\{t_{s1}, t_{s2}, \dots, t_{sy_s}\}$ be the times for the first cues emanated from y_s individuals perceived at the s -th survey location.

- Individuals who do not emanate at least one cue before time t_s will not be detected. Hence the actual number of individuals in the survey area is larger than or equal to y_s .
- Let the survey area be A_s .

1.1.1 Estimating the cue rate

It is known that time to the first cue is distributed as an Exponential distribution with rate δ . We observe the process for a finite time interval $(0, T]$. The corresponding distribution is, thus, a truncated Exponential distribution. Hence we can estimate δ by maximizing the likelihood function:

$$L(\delta; \underline{t}) = \prod_{s=1}^S \prod_{i=1}^{y_s} \{\delta \exp(-\delta t_{si}) / (1 - \exp(-\delta T))\} \quad (1)$$

Let us denote the maximum likelihood estimator by $\hat{\delta}$.

1.1.2 Estimating the population density

- Under the assumptions, the true number of individuals N_s (unknown to the observer) at the survey locations follow Poisson distribution with mean DA_s for $s = 1, 2, \dots, S$. That is, $N_s \sim \text{Poisson}(DA_s)$.
- Assume that individuals are detected independent of each other.
- Probability of detection for an individual is the same as the probability that an individual emanates at least one cue.
- It follows that, given $N_s = n_s$, the observed number of birds $Y_s | N_s = n_s \sim \text{Binomial}(n_s, (1 - \exp(-\delta t_s)))$. Let us use the notation $p_s(\delta) = 1 - \exp(-\delta t_s)$.
- We can write the likelihood for (D, δ) using the Binomial-Poisson mixture model:

$$L(D, \delta; \underline{y}_s, \underline{t}_s, \underline{A}_s) = \prod_{s=1}^S \sum_{n_s=y_s}^{\infty} \binom{n_s}{y_s} p_s(\delta)^{y_s} (1-p_s(\delta))^{n_s-y_s} e^{-DA_s} \frac{(DA_s)^{n_s}}{n_s!} \quad (2)$$

- We have an independent estimate of the cue rate $\delta = \hat{\delta}$ from equation (1). To estimate D , we maximize the pseudo-likelihood function (Gong and Samaniego, 1981):

$$L(D, \hat{\delta}; \underline{y}_s, \underline{t}_s, \underline{A}_s) = \prod_{s=1}^S \sum_{n_s=y_s}^{\infty} \binom{n_s}{y_s} p_s(\hat{\delta})^{y_s} (1-p_s(\hat{\delta}))^{n_s-y_s} e^{-DA_s} \frac{(DA_s)^{n_s}}{n_s!} \quad (3)$$

where $p_s(\hat{\delta}) = (1 - \exp(-\hat{\delta} t_s))$.

- We note that standard error for $\hat{D}$ cannot be based on the Hessian matrix for the pseudo-likelihood in equation (3) (Gong and Samaniego, 1981). Such a standard error underestimates the uncertainty. One can use non-parametric bootstrap to estimate the standard error and confidence interval.
- We can account for the effect of covariates on the density by using a log-linear model $D_s = \exp(X_s\beta)$.

1.2 Distance-dependent perception of cues

In practice, individuals are located at different distances from the observer. A cue from an individual that is farther from the observer is less likely to be perceived than one that is closer. We can modify the above procedure to take into account the distance effect on the perception of a cue. In addition to the assumptions made earlier, we assume that

- Birds do not move in or out of the spatial buffer during the survey duration.
- An individual that is at distance r , given that it emanates a cue, is detected by the observer with probability $\nu(r, \varphi)$ where r denotes the distance from the observer. Following the distance sampling ideas, we assume that $\nu(r, \varphi)$ is a decreasing function such that $\nu(0, \varphi) = 1$. The condition that $\nu(0, \varphi) = 1$ is not strictly necessary for the identifiability of D . It is, however, necessary to identify δ .
- The data are $\{(t_{s1}, r_{s1}), (t_{s2}, r_{s2}), \dots, (t_{sy_s}, r_{sy_s})\}$, the times for the cues along with their distance from the observer for $Y_s = y_s$ individuals at the s -th survey location where $s = 1, 2, \dots, S$.

1.2.1 Estimating the cue rate and distance effect parameters

The observed temporal point process of the cues in the presence of the distance effect is a thinned Poisson point process with thinning function $\nu(r, \varphi)$. Hence the observed process has the rate function $\delta(r, \delta_0, \varphi) = \delta_0 \nu(r, \varphi)$ where δ_0 is the innate cue rate as in the formulation in Section 1.1 above. The likelihood function for the observed temporal process is given as

$$L(\delta_0, \varphi; (\underline{t}, \underline{r})) = \prod_{s=1}^S \prod_{i=1}^{y_s} \{\delta_0 \nu(r_{si}, \varphi) \exp(-\delta_0 \nu(r_{si}, \varphi) t_{si}) / (1 - \exp(-\delta_0 \nu(r_{si}, \varphi) T))\} \quad (4)$$

We can estimate (δ_0, φ) by maximizing this likelihood function. Let us denote the estimate by $(\hat{\delta}_0, \hat{\varphi})$.

Probability of detecting an individual that is at distance r within the survey duration of t_s is given by

$$p(\delta_0, \varphi, r, t_s) = 1 - \exp(-\delta_0 \nu(r, \varphi) t_s)$$

and its estimate is

$$p(\hat{\delta}_0, \hat{\varphi}, r, t_s) = 1 - \exp(-\hat{\delta}_0 \nu(r, \hat{\varphi}) t_s)$$

1.2.2 Estimating the density

When detection of a cue depends on the distance to the observer, each individual has a different probability of detection. In this situation, the distribution of Y_s , the number of individuals observed at a survey location, does not follow a Binomial distribution. The distribution of Y_s , in fact, is a convolution of independent Bernoulli random variables.

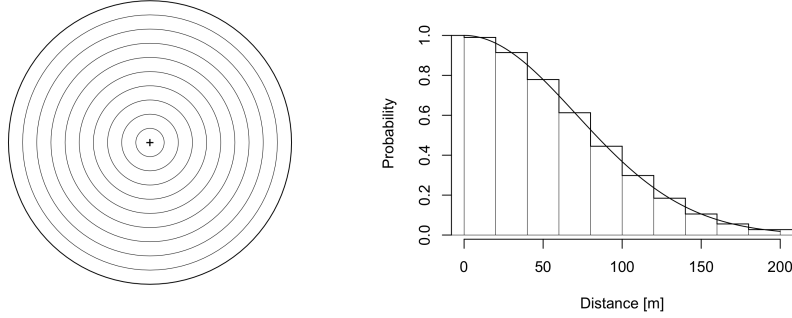
Suppose there are y_s individuals present in a survey location. Let the true distances at which these individuals are located be $\{r_{1s}, r_{2s}, \dots, r_{y_s s}\}$. The probabilities of detection for these individuals are $\{p(r_{is}, \delta_0, \varphi), i = 1, 2, \dots, y_s\}$. These are not identical to each other. We can write $Y_s = \sum_{i=1}^{y_s} Y_i$ where $Y_i \sim \text{Bernoulli}(1, p(r_{is}, \delta_0, \varphi))$. The random variable Y_s is a convolution of independent Bernoulli random variables.

It is extremely difficult to write the probability mass function (pmf) for this random variable. There are various approximations to the distribution of Y_s (Liu and Quertermos (2017)). One of the simplest approximations is to use $\text{Binomial}(N, \bar{p})$ where $\bar{p} = \frac{1}{N} \sum_{i=1}^{n_s} p_i$. Another possibility is to use the median or another measure of central tendency. In general, these approximations are reasonable if p_i 's do not vary too much.

Next we describe an approximation to the likelihood function when information on the time of the first cue and distance from the observer are known. We call this the Convolution likelihood although for mathematical convenience we are actually using a Binomial approximation to the Convolution distribution.

- Let us consider the s -th survey location.
- Without loss of generality, let $(0, 0)$ denote the location of the observer.
- Consider the concentric circles of the radii $\{0 < d_1 < d_2 < \dots < r_s\}$ around the location of the observer. We consider J such circles such that the largest circle encompasses the survey area of interest.
- If the increment in the radius of the concentric circles is small, probability of detection for all individuals who are in the donut is nearly identical. See figure A1.
- Hence we can use the Binomial approximation to the convolution distribution for the detection process of the individuals within each of the concentric donuts.
- The area of the first circle is $a_1 = \pi d_1^2$, area between the first circle and the second circle is $a_2 = \pi d_2^2 - a_1$ and so on. We divide the survey area in disjoint sets of areas $\{a_1, a_2, \dots, a_J\}$.

- The data for any specific survey location can be reformatted as counts in each of the donuts in figure A1. We denote the data as $\{(y_{sj}, d_j), j = 1, 2, \dots, J\}$. Notice that $y_s = \sum_{j=1}^J y_{sj}$.
- Let us look at the first circle. Probability of detection within this circle varies from 1 to $1 - \exp(-\delta_0 \nu(d_1, \gamma) t_s)$. Similarly probability of detection for the area between the first and the second circle varies from $1 - \exp(-\delta_0 \nu(d_1, \gamma) t_s)$ to $1 - \exp(-\delta_0 \nu(d_2, \gamma) t_s)$ and so on. One can approximate the probability of detection by using a step function where we replace $p(\delta_0, \gamma, d, t_s) \approx p(\delta_0, \gamma, 0.5(d_j - d_{j-1}), t_s)$ when $d_{j-1} \leq d < d_j$ and $d_0 = 0$ and $d_J = r_s$. This step function approximation is good provided $\max_{1 \leq j \leq J} (d_j - d_{j-1})$ is small. See Figure below.



- Let us look at the counts in the first circle for the s -th survey location. Under the step function approximation, the probability of detection within this circle is nearly constant. Hence the observed count in this circle can be modelled by a Binomial-Poisson mixture model. Its likelihood can be written as:

$$L_{s1}(D, \hat{\delta}_0, \hat{\gamma}; y_{s1}, d_1, t_s, a_1) = \sum_{N_1=y_{s1}}^{\infty} \binom{N_1}{y_{s1}} p(\hat{\delta}_0, \hat{\gamma}, 0.5d_1)^{y_{s1}} (1 - p(\hat{\delta}_0, \hat{\gamma}, 0.5d_1))^{N_1 - y_{s1}} e^{-Da_1} \frac{(Da_1)^{N_1}}{N_1!} \quad (5)$$

Now let us look at the likelihood function for the count in the area between

the first and the second circle:

$$L_{s2}(D, \hat{\delta}_0, \hat{\gamma}; y_{s2}, d_2, t_s, a_j) = \sum_{N_2=y_{s2}}^{\infty} \binom{N_2}{y_{s2}} p(\hat{\delta}_0, \hat{\gamma}, 0.5(d_2 - d_1))^{y_{s2}} (1 - p(\hat{\delta}_0, \hat{\gamma}, 0.5(d_2 - d_1)))^{N_2 - y_{s2}} e^{-D(a_2 - a_1)} \frac{(D(a_2 - a_1))^{N_2}}{N_2!} \quad (6)$$

We can follow this logic for other concentric circles and write

$$L_{sj}(D, \hat{\delta}_0, \hat{\gamma}; y_{sj}, d_j, t_s, a_j) = \sum_{N_j=y_{sj}}^{\infty} \binom{N_j}{y_{sj}} p(\hat{\delta}_0, \hat{\gamma}, 0.5(d_j - d_{j-1}))^{y_{sj}} (1 - p(\hat{\delta}_0, \hat{\gamma}, 0.5(d_j - d_{j-1})))^{N_j - y_{sj}} e^{-D(a_j - a_{j-1})} \frac{(D(a_j - a_{j-1}))^{N_j}}{N_j!} \quad (7)$$

The likelihood function corresponding to one of the survey sites is a product of all these likelihoods

$$L_s(D, \hat{\delta}_0, \hat{\gamma}) = \prod_{j=1}^J L_{sj}(D, \hat{\delta}_0, \hat{\gamma})$$

And the likelihood function for the entire data set is simply:

$$L(D, \hat{\delta}_0, \hat{\gamma}) = \prod_{s=1}^S \prod_{j=1}^J L_{sj}(D, \hat{\delta}_0, \hat{\gamma})$$

Our “gold standard” estimator is obtained by maximizing this likelihood function.

1.3 Likelihood functions used in SQPAD

In the following, we derive the likelihood function used to estimate the density D using the SQPAD. Unlike in Section 1.2, we do not require information on the distances to the individual birds nor the first time these individuals emanated a cue. The data are $\{(y_s, t_s, r_s), s = 1, 2, \dots, S\}$ where y_s is the number of distinct individuals detected at the survey location with spatial buffer of radius r_s and survey duration t_s .

With this limited information, we can not use the likelihood function in Section 1.2.2. Nor can we estimate the cue rate δ and the visibility bias parameter φ first and then estimate the density parameter D as was done in Section 1.2. However, we can estimate the entire parameter vector (D, δ, φ) simultaneously using the following approximation to full information likelihood function described in Section 1.2.2. Simulations show that this approximation is good.

1.3.1 SQPAD-C: count data

- Under the (spatial) HPP assumption, we know that the probability of N_s events in the area $A_s = \pi r_s^2$ is $e^{-DA_s} \frac{(DA_s)^{N_s}}{N_s!}$.
- The perceived counting process for the cues for an individual who is located at distance r is a thinned THPP with rate $\delta v(r; \varphi)$. Hence an individual who is located at distance r will be perceived if it emanates at least one cue before time t_s and the probability that it is perceived is $(1 - \exp(-\delta v(r; \varphi)t_s))$. However each individual is at a different distance from the observer. Hence the probability of detection is different for each individual.
- As before, $Y_s|N_s = y_s$ does not follow a Binomial distribution. But we can approximate it using a Binomial distribution as was discussed in Section 1.2.2.
- We replace each individual detection probabilities by $p_s = (1 - \exp(-\delta v(0.5r_s; \varphi)t_s))$.
- The average distance from the observer to an event within the buffer of radius r_s is $2/3 \approx 0.67r_s$. We can also use $p_s = 1 - \exp(-\delta v(0.67r_s; \gamma)t_s)$.
- We can also use the average probability of detection within the spatial buffer. This average probability can be computed for some standard visibility bias functions (Buckland et al. 2004, Sólomos et al. 2013). However for a general visibility function computing the average probability of detection can be difficult. Whereas evaluating the visibility function at a specific point is easy. At least under our simulations, all these approximations lead to similar results. These comparisons are not reported.

Using this approximation, we can write the likelihood function under the Binomial-Poisson mixture distribution as:

$$L(\theta; data) = \prod_{s=1}^S \sum_{N_s=y_s}^{N_{max}} e^{-DA_s} \frac{(DA_s)^{N_s}}{N_s!} \binom{N_s}{y_s} p(t_s, r_s; \delta_0, \tau)^{y_s} (1 - p(t_s, r_s; \delta_0, \tau))^{N_s - y_s}$$

where N_{max} is the maximum value of the abundance (strictly speaking this is ∞).

1.3.2 SQPAD-Occupancy data:

The data are the same as above except all Y_s 's that are 1 or more are transformed to 1. This is a censored data. The likelihood function for the occupancy data can be split into two components

$$L(\theta; data) = L_0(\theta; data) L_1(\theta; data)$$

Component 1: We observe 0 individuals because there are 0 individuals or if there are j individuals, we miss all of them.

$$L_0(\theta; data) = \prod_{s=1}^{S_0} \sum_{N_s=0}^{N_{max}} e^{-DA_s} \frac{(DA_s)^{N_s}}{N_s!} (1 - p(t_s, r_s; \delta_0, \tau))^{N_s}$$

Component 2: We observe at least 1 individual if there is only 1 individual and we observe it, or if there are $j > 1$ individuals, we observe either 1 or 2 or 3 etc. up to j (all of them) but censor it to 1.

$$L_1(\theta; data) = \prod_{s=1}^{S_1} \sum_{N_s=1}^{N_{max}} \sum_{j=1}^{N_s} e^{-DA_s} \frac{(DA_s)^{N_s}}{N_s!} \binom{N_s}{j} p(t_s, r_s; \delta_0, \tau)^j (1 - p(t_s, r_s; \delta_0, \tau))^{N_s-j}$$

We note here that these likelihood functions are identical to ones that were used in Lele et al. (2012) for the single visit occupancy data and Sólomos et al. (2012) for the single visit count data. The only difference is that we are explicitly using duration of the survey (t_s) and radius of the survey area (r_s) as covariates to model the probability of detection. These covariates and the fact that they vary from location to location satisfies the conditions for identifiability of the detection probability without resorting to multiple visits and close population assumption.

1.3.3 SQPAD-Censored count data

In this case, we believe that, in practice, we can confidently identify individuals if the total number of individuals is either 2 or 3. But this additional information could possibly improve the performance. The likelihood function in this case is split into $M + 1$ components. In the following, S_1 is a set of all locations where 1 individual was detected, S_2 is a set of all locations where 2 individuals were detected and so on. The last set S_M is of those locations where M individuals were detected.

$$L_0(\theta; data) = \prod_{s=1}^{S_0} \sum_{N_s=0}^{N_{max}} e^{-DA_s} \frac{(DA_s)^{N_s}}{N_s!} (1 - p(t_s, r_s; \delta_0, \tau))^{N_s}$$

$$L_1(\theta; data) = \prod_{s=1}^{S_1} \sum_{N_s=1}^{N_{max}} e^{-DA_s} \frac{(DA_s)^{N_s}}{N_s!} \binom{N_s}{1} p(t_s, r_s; \delta_0, \tau)^1 (1 - p(t_s, r_s; \delta_0, \tau))^{N_s-1}$$

$$L_2(\theta; data) = \prod_{i=1}^{S_2} \sum_{N_s=2}^{N_{max}} e^{-DA_s} \frac{(DA_s)^{N_s}}{N_s!} \binom{N_s}{2} p(t_s, r_s; \delta_0, \tau)^2 (1 - p(t_s, r_s; \delta_0, \tau))^{N_s-2}$$

$\vdots$

$$L_M(\theta; data) = \prod_{s=1}^{S_M} \sum_{N_s=M}^{N_{max}} e^{-DA_s} \frac{(DA_s)^{N_s}}{N_s!} \left\{ \sum_{j=M}^{N_s} \binom{N_s}{j} p(t_s, r_s; \delta_0, \tau)^j (1 - p(t_s, r_s; \delta_0, \tau))^{N_s-j} \right\}$$

Full likelihood function is a product of all these components.

1.4 Composite likelihood for analyzing data from QPAD protocol

If the data are collected using the sampling protocol appropriate for QPAD, we use the composite likelihood function as follows.

1. For each site, we construct a new dataset that consists of total count for all possible bins of the type $(0, t_j) \times (0, r_k)$. For example, if the original data are for time intervals $(0, 3)$, $(3, 6)$ and distance bins $(0, 50) \times (50, 100)$, we compute the total counts in the bins $(0, 3) \times (0, 50)$, $(0, 3) \times (0, 100)$, $(0, 6) \times (0, 50)$ and $(0, 6) \times (0, 100)$.
2. We then compute the likelihood for each of these bins using the likelihood formula for SQPAD-count or SQPAD-occupancy.
3. The product of all these likelihood components is known as the composite likelihood function. It is known (Lindsay 1988) that the maximum composite likelihood estimator is a consistent estimator. The standard error and confidence intervals can be computed using nonparametric bootstrap as described in the main text.

1.5 Identifiability of the parameters

It may seem surprising that the density parameter D as well as the cue rate δ and distance effect parameter φ can be estimated using SQPAD. There are two features of SQPAD that make this possible.

1. The survey duration t_s and survey area (radius) r_s vary from location to location.
2. Probability of detection is 0 when the survey duration t_s is 0 and it *monotonically* increases to 1 as $t_s \rightarrow \infty$. Furthermore, we assume that the probability of detection is a continuous function of time.

First condition is completely under the control of the researcher and can be satisfied by design. The second condition, in practical terms, says that the longer one monitors a survey location, higher is the probability of detecting an individual (if it is present) and that if an individual is present, eventually it will be detected. This condition is satisfied if the cue process is a renewal process with independent increments. (Notice that the Removal sampling method also assumes that the cue process is a renewal process with independent increments.)

We emphasize that the renewal process assumption is not critical; what is critical is that the probability of detection *monotonically* increases from 0 to 1 as a function of time. It may be satisfied even when the underlying cue process is not a renewal process.

For pedagogical reasons, let us start with the case where distance does not affect the perception of a cue and that the spatial buffer is the same from location to location. Hence the survey area is A for all locations. In this case, we know that an individual will be detected during an interval $(0, t]$ is given by: $p(t; \delta) = 1 - \exp(-\delta t)$. We consider the following hierarchical model:

Hierarchy 1 (Process model): $N_s \sim \text{Poisson}(DA)$

Hierarchy 2 (Observation model): $Y_s | N_s = n_s \sim \text{Binomial}(n_s, p(t_s; \delta))$

The marginal mean of Y_s can be computed as: $E(Y_s) = E(E(Y_s | N_s)) = p(t_s; \delta)DA$. This is a function of duration t . To prove the identifiability of the parameters, it is sufficient (though not necessary) to show that: $p(t; \delta_1)D_1A = p(t; \delta_2)D_2A$ if and only if $(D_1 = D_2, \delta_1 = \delta_2)$ for all $t > 0$

1. If $(D_1 = D_2, \delta_1 = \delta_2)$, it is obvious that $p(t; \delta_1)D_1A = p(t; \delta_1)D_1A$ for all $t > 0$.
2. To show the converse, we will use the method of *reductio ad absurdum*.

For some $D_1 \neq D_2, \delta_1 \neq \delta_2$, let $p(t; \delta_1)D_1A = p(t; \delta_2)D_2A$ for all $t > 0$.

$$\Rightarrow \frac{D_1}{D_2} = \frac{p(t; \delta_2)}{p(t; \delta_1)} \text{ for all } t > 0$$

$$\Rightarrow k = \frac{(1 - \exp(-\delta_2 t))}{(1 - \exp(-\delta_1 t))} \text{ for all } t > 0 \text{ and } k = \frac{D_1}{D_2} \neq 1.$$

This is impossible because the right hand side converges to 1 as $t \rightarrow \infty$ and left hand side is strictly different than 1. We have reached an absurdity!

Hence $p(t; \delta_1)D_1A = p(t; \delta_2)D_2A$ if and only if $(D_1 = D_2, \delta_1 = \delta_2)$ for all $t > 0$.

This proof can be extended to prove identifiability of the distance effect parameter φ when perception depends on distance in a similar manner.

Furthermore,

1. The RSPF condition (Lele and Keim 2006, Solymos and Lele 2016) simply demands that the ratio of two detection probability functions should not be a constant. This is satisfied as long as probability of detection is a continuous, monotone increasing function of some covariate that increases from 0 to 1. In the case of SQPAD, survey duration is a naturally available covariate that affects the probability of detection in this fashion.
2. If one is not interested in estimating cue rate δ and distance effect parameter φ separately, one need not vary the survey radius from location to location. As long as the survey duration varies from location to location, the density parameter D is identifiable.

We want to reiterate, that the novelty of the SQPAD protocol is that it explicitly incorporates survey area and duration as covariates in detection probability.

Varying the survey area and duration is completely under the control of the observer and this variable effort brings the variability in detection probabilities that is required for parameter identifiability.